CyberPulse Setup Walkthrough

Clone, configure and run the Security Awareness Quiz Platform on a Kali Linux VM, up to the signed-in dashboard. Database seeding is covered separately.

What you'll need

- A running Kali Linux VM with internet access (any Linux distro works, commands assume Debian/Ubuntu/Kali).
- Access to the **CyArt-Tech/Security-Awareness-Quiz-Platform** GitHub repository.
- A GitHub account you can sign in to in a browser.
- A Supabase account (free tier is fine) — sign up at supabase.com.
- Around 30 minutes.

Step 1 — Install base tools on the VM

Open a terminal and install git, curl and Node.js 22.

```
sudo apt update
sudo apt install -y git curl
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo bash -
sudo apt install -y nodejs
```

Verify Node is v22 or higher:

```
node -v
```

Note: Supabase's SDK requires Node 22+. Older Kali images ship with Node 18 or 20, which will install but print engine warnings. Always run the NodeSource step above.

Step 2 — Set up SSH access to GitHub

The repository is private, so you'll authenticate with an SSH key. This is simpler than using Personal Access Tokens.

2.1 Generate a key

Run this command, then press Enter at every prompt (no passphrase):

```
ssh-keygen -t ed25519
```

2.2 Copy the public key

```
cat ~/.ssh/id_ed25519.pub
```

The output is one long line starting with `ssh-ed25519`. Select the whole line and copy it with **Ctrl+Shift+C**.

2.3 Add the key to GitHub

- Open https://github.com/settings/keys in a browser.
- Click **New SSH key**.
- Title: anything (e.g. `kali-vm`).
- Key type: **Authentication Key** (the default).
- Paste the line into the Key box and click **Add SSH key**.

> **Access check:** The user must already be a collaborator on the **CyArt-Tech/Security-Awareness-Quiz-Platform** repo. If not, ask the repo owner to add them before step 3.

Step 3 — Clone the repository

```
cd ~
git clone git@github.com:CyArt-Tech/Security-Awareness-Quiz-Platform.git
cd Security-Awareness-Quiz-Platform
```

If asked *"Are you sure you want to continue connecting?"*, type **yes** and press Enter. A successful clone ends with `Resolving deltas: 100% done`.

> **Note:** If git says *"Repository not found"*, the user's GitHub account doesn't have access. Verify the SSH key was added correctly and repo access was granted.

Step 4 — Install project dependencies

```
npm install
```

This will download around 400 packages and takes 1–2 minutes. Warnings about deprecated packages are normal — ignore them. **Do not** run `npm audit fix --force`; it can break the project.

Step 5 — Create a Supabase project

The app uses Supabase for authentication and the database. Create a free project to get your API credentials.

5.1 Create the project

- Open https://supabase.com and sign in (GitHub login works).
- Click **New project**.
- Name it anything (e.g. `cyberpulse`).
- Set a database password and **save it somewhere safe**.
- Pick the region closest to you.
- Click **Create new project** and wait about 2 minutes until it says **Healthy**.

5.2 Collect the API credentials

Once the project shows Healthy, open **Project Settings** → **API Keys** (gear icon in the left sidebar). You need three values:

- **Project URL** — looks like `https://<id>.supabase.co`.
- **Publishable key** — starts with `sb_publishable_`.
- **Secret key** — starts with `sb_secret_`. Click **Reveal** to see it.

> **Legacy key names:** If the API Keys page shows "anon" and "service_role" keys instead, use **anon** as the publishable key and **service_role** as the secret key.

> **Security: Never share the secret key.** It bypasses all database security. Keep it out of chats, screenshots and git commits.

Step 6 — Configure environment variables

The app reads its Supabase credentials from a file called `.env.local`. Create it from the example file:

```
cd ~/Security-Awareness-Quiz-Platform
cp .env.example .env.local
nano .env.local
```

Edit the file so the three lines look like this — replace the placeholders with your real values from Step 5.2:

```
NEXT_PUBLIC_SUPABASE_URL=https://YOUR-PROJECT-ID.supabase.co
NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY=sb_publishable_...
SUPABASE_SECRET_KEY=sb_secret_...
```

Save and exit nano:

- **Ctrl+O**, then **Enter** to confirm the filename.
- **Ctrl+X** to exit.

> **Formatting:** No spaces around the **=** signs, no quotes around the values. A space will break the connection and the app will fail at startup.

Step 7 — Start the dev server

```
npm run dev
```

Wait until the terminal prints `Ready in ... ms` and shows the local URL. The server stays running — leave this terminal open.

> **Stop / restart:** To stop the server, press **Ctrl+C** in that terminal. To restart it later: `cd ~/Security-Awareness-Quiz-Platform && npm run dev`.

Step 8 — Open the app and sign in

Open Firefox on the VM and go to:

```
http://localhost:3000
```

8.1 Create an account

- Click **Create Account**.
- Enter an email and password.
- Supabase may send a confirmation email — click the link in the inbox if asked.

8.2 Confirm the dashboard loads

After signing in, you should see the **CyberPulse** dashboard with:

- The sidebar: Dashboard, Quiz Library, My Progress, Leaderboard, Achievements, Attempt History, Profile & Settings.
- Stat cards showing 0 completed, 0 points — this is expected before the database is seeded.
- Ten domain cards showing "Pending" status.

> **Expected state:** If the dashboard loads but shows no domains and empty stats, that's correct for this stage. The next walkthrough covers seeding the database so quizzes appear.

Troubleshooting

Git clone fails with "Repository not found"

The GitHub account doesn't have access to the repo, or the SSH key wasn't added. Re-check `cat ~/.ssh/id_ed25519.pub` matches what's listed at github.com/settings/keys, and that the user is a collaborator on CyArt-Tech/Security-Awareness-Quiz-Platform.

npm install fails with an engine warning

Node is too old. Re-run the NodeSource step from Step 1 and verify `node -v` prints v22 or higher. Then: `rm -rf node_modules && npm install`.

Browser shows "Invalid API key" or "Failed to fetch"

The `.env.local` values are wrong. Check for extra spaces, quotes or a trailing newline. Stop the server (Ctrl+C), fix the file, and run `npm run dev` again — env changes aren't picked up live.

Firefox freezes the VM

Not enough resources. In UTM: shut the VM down, open the VM settings, raise Memory to at least 8192 MiB and CPU Cores to 4. Also check **Display** → enable hardware acceleration if offered.

Port 3000 is already in use

An old dev server is still running. Kill it: `pkill -f 'next dev'`, then `npm run dev` again.

Next up

At this point the app is running and the user is signed in, but the database has no tables yet. The next walkthrough covers:

- Running the schema and seed SQL in Supabase's SQL Editor.
- Promoting the user account to admin.
- Verifying the 10 security domains and quizzes appear on the dashboard.